

Section 5: Interpreting categorical and count models — worked solutions

GOVT 6039 · Lecture 11 practice

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Worked solutions.

The three cases use simulated data. Round numerical answers to two or three decimal places.

Focus: cumulative odds versus category probabilities; multinomial log odds versus probability changes; count ratios versus count differences; and what overdispersion changes.

Choose the outcome model

Choose **ordered logit**, **multinomial logit**, or a **count model** for each outcome, and give one reason: (1) trust in city government: low, medium, high; (2) one intended action: no action, petition, or rally; (3) number of protests in a district during one year.

💡 Worked solution

1. **Trust: ordered logit.** We know $\text{Low} < \text{Medium} < \text{High}$, but we do not know whether the Low-to-Medium gap equals the Medium-to-High gap. Ordered logit uses the ranking without treating the labels as equally spaced numbers.
2. **Participation: multinomial logit.** No action, Petition, and Rally are different choices, but there is no single numerical ranking that all three must follow. A multinomial model estimates the probability of each choice, using one as the reference.
3. **Protests: a count model.** The outcome can be 0, 1, 2, and so on. A count model respects that lower bound and can relate predictors to the expected number of protests. The Poisson model is a starting point; its mean-variance restriction should be checked.

Coding trust as 1, 2, 3 and using OLS would impose equal spacing. Turning the protest total into any/none would discard how many protests occurred.

A. Ordered trust

A survey asks residents to rate trust in city government as **Low** < **Medium** < **High**. **visit** = 1 means a resident attended a city information session; **engagement** is a standardized prior civic-engagement score. The fitted ordered logit uses

$$P(Y \leq j \mid x) = \text{logit}^{-1}(\zeta_j - x^\top \beta).$$

The coefficient on **visit** is **0.862**; the coefficient on **engagement** is **0.618**. Estimated cutpoints are **-0.548** and **0.953**. The cutpoints absorb the intercept.

Table 1: Predicted category probabilities for four residents

Engagement	Visit	Low	Medium	High
-1	0	0.517	0.310	0.172
-1	1	0.312	0.358	0.330
+1	0	0.238	0.345	0.417

Engagement	Visit	Low	Medium	High
+1	1	0.116	0.255	0.629

1. At fixed engagement, what does the positive **visit** coefficient imply for the odds of **High** rather than **Low** or **Medium**? Compute the odds ratio.
2. At engagement = -1, calculate the visit contrast for **each** category's probability. Repeat for Medium at engagement = +1. Does a positive coefficient guarantee that the probability of Medium rises?
3. Why do the three probability changes at any given engagement sum to zero? Explain why the coefficient is **not** the change in $P(\text{High})$.
4. What is held fixed? Does the fitted contrast, by itself, establish a causal effect of attending the session?

💡 Worked solution

1. Derive the odds ratio. Let v denote visit and e denote engagement. Set $\eta(v, e) = \hat{\beta}_v v + \hat{\beta}_e e$ and $C_j(v, e) = P(Y \leq j | v, e) = F(\hat{\zeta}_j - \eta(v, e))$, where $F(t) = (1 + e^{-t})^{-1}$. Logistic odds give

$$\log \frac{P(Y > j | v, e)}{P(Y \leq j | v, e)} = \eta(v, e) - \hat{\zeta}_j, \quad j = 1, 2.$$

Holding e fixed and changing $v : 0 \rightarrow 1$ therefore adds $\hat{\beta}_v = 0.862$ to *both* cumulative log-odds comparisons. For $j = 2$, the comparison is High versus Low or Medium:

$$\text{OR} = \exp(\hat{\beta}_v) = e^{0.862} = 2.369.$$

The inverse odds, at or below j versus above j , have ratio $e^{-0.862} = 0.422$.

2. Recover category probabilities before taking differences. With three ordered categories,

$$p_{\text{Low}} = C_1, \quad p_{\text{Medium}} = C_2 - C_1, \quad p_{\text{High}} = 1 - C_2.$$

At engagement $e = -1$, $\eta(0, -1) = -0.618$ and $\eta(1, -1) = 0.245$. Substituting the cutpoints into F gives

$$[C_1(0, -1), C_2(0, -1)] = (0.517, 0.828), \quad [C_1(1, -1), C_2(1, -1)] = (0.312, 0.670).$$

Thus the fitted probability vectors are (0.517, 0.310, 0.172) without the visit and (0.312, 0.358, 0.330) with it. Subtracting component by component yields

$$\Delta p(-1) = (-0.206, +0.048, +0.158).$$

At $e = +1$, the Medium probability changes from 0.345 to 0.255, or **-0.091**. Its sign reverses across these two engagement values.

3. Check the adding-up restriction. For either visit status, $p_{\text{Low}} + p_{\text{Medium}} + p_{\text{High}} = 1$. Hence

$$\Delta p_{\text{Low}} + \Delta p_{\text{Medium}} + \Delta p_{\text{High}} = 1 - 1 = 0.$$

The coefficient belongs to the cumulative *log-odds* equation; it cannot be read as a change in $P(\text{High})$.

4. State the comparison and its limit. These calculations hold engagement fixed while changing visit status. The fitted model alone does not establish that attendance was independent of other causes of trust, so the numerical contrast is not, by itself, a causal effect.

B. Unordered participation choice

Residents choose **one** response to a proposed permit rule: No action, Petition, or Rally. **permit = 1** denotes the new-rule setting. **No action** is the base category.

Table 2: Multinomial logit coefficients

Outcome_vs_No_action	Intercept	Permit
Petition	0.419	-0.499

Outcome_vs_No_action	Intercept	Permit
Rally	0.509	-1.441

Table 3: Predicted probabilities at each permit setting

Permit	No_action	Petition	Rally
0	0.239	0.363	0.398
1	0.432	0.398	0.170

1. Translate the **permit** coefficient for Petition into a sentence about **Petition versus No action** odds. Calculate the ratio.
2. Does the **Petition probability** move in the same direction as its coefficient? Explain using all three outcomes.
3. Give the probability change for each outcome. Check that the three changes sum to zero.
4. If Petition and Rally were nearly identical substitutes, which assumption of multinomial logit would deserve scrutiny? State the assumption using an odds ratio.

💡 Worked solution

1. Define the baseline comparison. Write r for the permit indicator, $s_0(r) = 1$ for No action, $s_P(r) = e^{\alpha_P + \beta_P r}$ for Petition, and $s_R(r) = e^{\alpha_R + \beta_R r}$ for Rally. The probability of option j is $p_j(r) = s_j(r) / [s_0(r) + s_P(r) + s_R(r)]$. Dividing Petition by No action cancels the common denominator:

$$\log \frac{p_P(r)}{p_0(r)} = \alpha_P + \beta_P r.$$

Changing $r : 0 \rightarrow 1$ multiplies these *relative odds* by $e^{\hat{\beta}_P} = e^{-0.499} = 0.607$. This is not a probability change.

2. Normalize all three scores to obtain probabilities. In the order (No action, Petition, Rally), the score vectors are

$$s(0) = (1, e^{0.419}, e^{0.509}) \approx (1.000, 1.520, 1.664), \quad \sum_j s_j(0) = 4.184;$$

$$s(1) = (1, e^{0.419-0.499}, e^{0.509-1.441}) \approx (1.000, 0.923, 0.394), \quad \sum_j s_j(1) = 2.317.$$

Dividing each score by its row sum gives $p(0) = (0.239, 0.363, 0.398)$ and $p(1) = (0.432, 0.398, 0.170)$. Petition's probability rises because Rally loses a larger share than Petition, even though Petition's odds relative to No action fall.

3. Take the probability contrasts. Subtract $p(0)$ from $p(1)$:

$$\Delta p = (+0.193, +0.035, -0.228), \quad \sum_j \Delta p_j = 0.$$

The sum is zero because the options are mutually exclusive and exhaustive.

4. State the IIA restriction. The model requires

$$\frac{p_P(r)}{p_0(r)} = \frac{s_P(r)}{s_0(r)} = e^{\alpha_P + \beta_P r},$$

which contains no Rally score. If Rally were removed, IIA would leave the Petition/No action odds unchanged. When Rally is a close substitute for Petition, its removal may instead send disproportionately many people to Petition. That is the substantive concern.

C. Protest counts

The outcome is the number of protests in a district-year. **outreach** = 1 indicates a civic outreach network; **pop_z** is standardized district population. Poisson, quasi-Poisson, and negative binomial (NB) models are fit to the same simulated observations. All three use $\log(\mu_i) = x_i^\top \beta$.

Table 4: Count-model estimates and standard errors

Term	Pois_beta	Pois_SE	Quasi_SE	NB_beta	NB_SE
Intercept	0.277	0.022	0.036	0.281	0.033
Outreach	0.633	0.027	0.044	0.628	0.044
Population (standardized)	0.313	0.013	0.021	0.310	0.022

At $\text{pop_z} = 0$, Poisson predicts **1.319** protests when $\text{outreach} = 0$ and **2.485** when $\text{outreach} = 1$. The quasi-Poisson dispersion estimate is **2.68**. The NB size parameter is **1.16**.

1. Interpret the Poisson outreach coefficient as a ratio of expected counts. Compute the ratio. Then calculate the predicted **count difference** at $\text{pop_z} = 0$.
2. Would the predicted count difference be the same in a more populous district? Explain without refitting.
3. Compare Poisson and quasi-Poisson estimates and standard errors. What does dispersion greater than one suggest? What does quasi-Poisson change?
4. For NB, write the conditional variance at mean $\mu = 2$ using the estimated θ . Compare it with the Poisson variance at the same mean. Can the NB outreach coefficient be interpreted with the same exponential rule?

💡 Worked solution

1. From log mean to count ratio. Write a for outreach and z for standardized population. The fitted Poisson mean is

$$\hat{\mu}(a, z) = \exp(0.277 + 0.633a + 0.313z).$$

Holding z fixed, divide the two fitted means:

$$\frac{\hat{\mu}(1, z)}{\hat{\mu}(0, z)} = \frac{e^{\hat{\beta}_0 + \hat{\beta}_a + \hat{\beta}_z z}}{e^{\hat{\beta}_0 + \hat{\beta}_z z}} = e^{\hat{\beta}_a} = e^{0.633} = 1.884.$$

At $z = 0$, $\hat{\mu}(0, 0) = 1.319$ and $\hat{\mu}(1, 0) = 2.485$, so the fitted count difference is $2.485 - 1.319 = 1.166$ protests.

2. Why the difference changes with population. Algebraically,

$$\Delta(z) = \hat{\mu}(1, z) - \hat{\mu}(0, z) = \hat{\mu}(0, z)(e^{\hat{\beta}_a} - 1).$$

The ratio $e^{\hat{\beta}_a}$ is fixed, but $\hat{\mu}(0, z)$ changes. At $z = 1$, the two means are 1.805 and 3.399; their difference is 1.595. Across the $n = 3000$ simulated districts, the average fitted difference is

$$\frac{1}{n} \sum_{i=1}^n [\hat{\mu}(1, z_i) - \hat{\mu}(0, z_i)] = 1.222.$$

3. Separate the mean assumption from the variance assumption. Poisson imposes $\text{Var}(Y_i | x_i) = \mu_i$. Quasi-Poisson instead permits $\text{Var}(Y_i | x_i) = \phi \mu_i$. The estimated $\hat{\phi} = 2.684$, so the quasi standard errors are approximately $\sqrt{\hat{\phi}} = 1.638$ times the ordinary Poisson values. For outreach, $0.027 \times 1.638 \approx 0.044$. The coefficient and the mean predictions stay the same. If the conditional mean and independence assumptions are right, Poisson coefficients can remain consistent despite overdispersion; its unadjusted standard errors do not.

4. Interpret the negative binomial fit. It retains a log mean but specifies

$$\text{Var}(Y_i | x_i) = \mu_i + \frac{\mu_i^2}{\theta}.$$

At $\mu_i = 2$ and $\hat{\theta} = 1.156$, the fitted variance is $2 + 4/1.156 = 5.460$, compared with the Poisson variance of 2. The NB outreach coefficient is 0.628, hence its count ratio is $e^{0.628} = 1.873$. The exponential interpretation survives because the mean still uses a log link; the NB coefficient can differ because the likelihood differs.

Check your interpretation

Answer each question briefly.

1. An ordered-logit coefficient is positive. Must the middle-category probability increase? Why?
2. A multinomial coefficient for Petition versus No action is negative. Must $P(\text{Petition})$ fall? Why?
3. A Poisson coefficient is 0.7. What remains constant across covariate profiles: the ratio or the difference in expected counts?

Worked solution

1. **No.** A positive ordered-logit coefficient shifts probability upward in the ordering, but $P(\text{Medium}) = P(Y \leq \text{Medium}) - P(Y \leq \text{Low})$. Both cumulative probabilities change. Their difference may rise or fall, as it did across the two engagement values above.
 2. **No.** The negative coefficient says that Petition/No action odds fall. To find $P(\text{Petition})$, divide Petition's score by the sum of all three scores. A large enough fall in Rally's score can make Petition's share rise.
 3. The **ratio** of expected counts remains $e^{0.7} \approx 2.01$ at every covariate profile in a log-link model without an interaction. The difference is $\mu_0(e^{0.7} - 1)$ and therefore grows with the baseline expected count μ_0 .
- Each answer hinges on keeping the coefficient's scale separate from a category probability or count difference.

Source and reproducibility

The R code in this QMD generates the data (`set.seed(6039)`) and fits the models with `MASS::polr`, `nnet::multinom`, `glm`, and `MASS::glm.nb`. Calculations use full-precision estimates, so the last digit may differ from arithmetic based on the printed tables. Topics and notation follow [Lecture 11: Catch-up on ordered, unordered and count outcomes](#).